

LAB 9

ANSWERS FOR PRACTICE QUESTIONS.

1) Generate random integer square matrices A of order 4, B of order 10, and C of order 15. Find their eigenvalues and eigenvectors.

a) Verify the following properties of eigenvalues for A, B and C

i) Trace of a matrix = sum of eigenvalues of the matrix.

```
% Generate random square matrices A, B, and C
```

```
A = randi([-10, 10], 4, 4);  
B = randi([-10, 10], 10, 10);  
C = randi([-10, 10], 15, 15);
```

```
% Find their eigenvalues and eigenvectors
```

```
[eigenvectors, eigenvalues] = eig(A);  
[eigenvectors, eigenvalues] = eig(B);  
[eigenvectors, eigenvalues] = eig(C);  
eigA = eig(A);  
eigB = eig(B);  
eigC = eig(C);
```

```
% Calculate trace and sum of eigenvalues
```

```
traceA = trace(A)
```

```
traceA =  
20
```

```
sumEigA = sum(eigA)
```

```
sumEigA =  
20.0000
```

```
traceB = trace(B)
```

```
traceB =  
-39
```

```
sumEigB = sum(eigB)
```

```
sumEigB =  
-39.0000
```

```
traceC = trace(C)
```

```
traceC =  
-1
```

```
sumEigC = sum(eigC)
```

```
sumEigC =  
-1.0000
```

```
% Check if trace equals sum of eigenvalues for all matrices.
if round(traceA) == round(sumEigA) && round(traceB) ==round(sumEigB) &&
round(traceC) == round(sumEigC)
    disp('Trace equals sum of eigenvalues for all matrices.');
```

```
else
    disp('Trace does not equal sum of eigenvalues for all matrices.');
```

```
end
```

```
Trace equals sum of eigenvalues for all matrices.
```

ii) Det of a matrix = Product of eigenvalues of the matrix.

```
% Calculate determinant as product of eigenvalues
D1 = det(A)
```

```
D1 =
-6.4980e+03
```

```
P1 = prod(eigA)
```

```
P1 =
-6.4980e+03 - 3.5190e-13i
```

```
D2 = det(B)
```

```
D2 =
-1.6930e+11
```

```
P2= prod(eigB)
```

```
P2 =
-1.6930e+11
```

```
D3 = det(C)
```

```
D3 =
3.8321e+17
```

```
P3=prod(eigC)
```

```
P3 =
3.8321e+17 - 3.2000e+01i
```

```
% Check if determinant equals product of eigenvalues for all matrices.
if round(D1) == round(P1) && round(D2) == round(P2) && round(D3) == round(P3)
    disp('Determinant equals product of eigenvalues for all matrices.');
```

```
else
```

```
    disp('Determinant does not equal product of eigenvalues for all matrices.');
```

```
end
```

```
Determinant does not equal product of eigenvalues for all matrices.
```

iii) Complex eigenvalues of a matrix appear in pairs.

```
eigenvalues_A = eig(A);
```

```
eigenvalues_B = eig(B);
eigenvalues_C = eig(C);

for M = {eigenvalues_A, eigenvalues_B, eigenvalues_C}
    if isequal(sort(conj(M{1}(imag(M{1})~=0))), sort(M{1}(imag(M{1})~=0)))
        disp('Complex eigenvalues appear in conjugate pairs.');
```

```
    else
        disp('Complex eigenvalues do NOT appear in conjugate pairs.');
```

b) Find eigenvalues of $5A$, B and $-C$. Compare them with eigenvalues of A , B and C respectively. What do you observe?

```
P= 5*A;
P1= eig(P)
```

```
P1 = 4x1 complex
-34.0704 + 0.0000i
 51.5585 +34.5359i
 51.5585 -34.5359i
 30.9534 + 0.0000i
```

```
Q=(B/3);
Q1=eig(Q)
```

```
Q1 = 10x1 complex
 4.9545 + 0.0000i
-7.0455 + 0.0000i
-4.0895 + 5.8558i
-4.0895 - 5.8558i
 1.1474 + 5.2114i
 1.1474 - 5.2114i
-1.2356 + 4.0456i
-1.2356 - 4.0456i
-1.2768 + 1.2368i
-1.2768 - 1.2368i
```

```
R=C^(-1);
R1=eig(R)
```

```
R1 = 15x1 complex
 0.0577 + 0.1019i
 0.0577 - 0.1019i
 0.0704 + 0.0000i
 0.0631 + 0.0173i
 0.0631 - 0.0173i
-0.0574 + 0.0701i
-0.0574 - 0.0701i
 0.0239 + 0.0446i
 0.0239 - 0.0446i
-0.0114 + 0.0637i
⋮
```

OBSERVATION: The eigen values are complex conjugate pairs.

c) Find eigenvectors of $\mathbf{P}$ and $\mathbf{Q}$. Compare them with eigenvectors of $\mathbf{A}$ and $\mathbf{B}$ respectively. What do you observe?

```
[eigVecP, ~] = eig(P)
```

eigVecP = 4×4 complex

```
-0.0231 + 0.0000i  -0.6425 + 0.0000i  -0.6425 + 0.0000i   0.3028 + 0.0000i
 0.8398 + 0.0000i  -0.4932 - 0.1093i  -0.4932 + 0.1093i   0.2149 + 0.0000i
 0.4088 + 0.0000i   0.0113 + 0.3759i   0.0113 - 0.3759i  -0.2213 + 0.0000i
 0.3566 + 0.0000i  -0.3912 + 0.1938i  -0.3912 - 0.1938i   0.9018 + 0.0000i
```

```
[eigVecQ, ~] = eig(Q)
```

eigVecQ = 10×10 complex

```
0.3782 + 0.0000i   0.7633 + 0.0000i  -0.0274 + 0.0525i  -0.0274 - 0.0525i ...
0.0288 + 0.0000i  -0.1916 + 0.0000i   0.0533 - 0.3091i   0.0533 + 0.3091i
0.1995 + 0.0000i  -0.1839 + 0.0000i  -0.5263 + 0.0000i  -0.5263 + 0.0000i
0.5035 + 0.0000i   0.0140 + 0.0000i   0.3748 - 0.3038i   0.3748 + 0.3038i
0.1301 + 0.0000i   0.1879 + 0.0000i  -0.0947 - 0.2472i  -0.0947 + 0.2472i
-0.3155 + 0.0000i -0.0725 + 0.0000i  -0.2619 - 0.2168i  -0.2619 + 0.2168i
-0.3263 + 0.0000i   0.1169 + 0.0000i  -0.0218 - 0.1018i  -0.0218 + 0.1018i
-0.4824 + 0.0000i   0.3603 + 0.0000i   0.0614 - 0.2975i   0.0614 + 0.2975i
0.3186 + 0.0000i  -0.1216 + 0.0000i  -0.2330 - 0.0287i  -0.2330 + 0.0287i
-0.0756 + 0.0000i   0.3844 + 0.0000i   0.0794 + 0.1953i   0.0794 - 0.1953i
```

```
[eigVecR, ~] = eig(R)
```

eigVecR = 15×15 complex

```
0.0635 + 0.0451i   0.0635 - 0.0451i   0.1222 + 0.0000i   0.1061 - 0.2380i ...
-0.1551 - 0.0447i  -0.1551 + 0.0447i   0.4800 + 0.0000i  -0.2043 - 0.2561i
-0.3162 + 0.2654i  -0.3162 - 0.2654i  -0.3564 + 0.0000i   0.3017 + 0.0685i
-0.0655 + 0.0079i  -0.0655 - 0.0079i   0.2775 + 0.0000i  -0.3003 - 0.0248i
-0.2005 - 0.0194i  -0.2005 + 0.0194i   0.0264 + 0.0000i  -0.0588 + 0.1142i
0.5187 + 0.0000i   0.5187 + 0.0000i   0.2921 + 0.0000i  -0.3309 + 0.0598i
-0.1922 - 0.0586i  -0.1922 + 0.0586i  -0.4512 + 0.0000i   0.3375 + 0.0000i
-0.1846 - 0.0241i  -0.1846 + 0.0241i  -0.1333 + 0.0000i   0.1751 - 0.0467i
-0.0934 - 0.0664i  -0.0934 + 0.0664i  -0.1754 + 0.0000i   0.2707 - 0.0213i
0.2092 + 0.1296i   0.2092 - 0.1296i   0.1042 + 0.0000i   0.0817 - 0.1374i
⋮
```

OBSERVATION: The eigenvectors appear in complex conjugate pairs.

d) From $\mathbf{A}$, using $\mathbf{S} = \mathbf{A} + \mathbf{A}'$ generate a symmetric matrix. Find the eigenvalues and eigenvectors of $\mathbf{S}$.

```
S=A+A'
```

S = 4×4

```
16    15     2     4
15     0   -17    -6
 2   -17    18    -5
 4    -6    -5     6
```

```
[eigenvectors, eigenvalues] = eig(S)
```

eigenvectors = 4×4

```
-0.3836    0.1219    0.7803   -0.4786
 0.7614   -0.3008    0.0718   -0.5698
```

```

0.4111    -0.0696    0.6204    0.6643
0.3227    0.9433   -0.0322   -0.0708
eigenvalues = 4x4
-19.2792         0         0         0
         0    8.7994         0         0
         0         0   18.8062         0
         0         0         0   31.6735

```

2) Generate a random integer square matrix A of order 3.

a) Find the eigenvalues, right eigenvectors, and left eigenvectors of this matrix.

```

clc
A = randi([-5,5],3);
[R,L]=eig(A)

```

```

R = 3x3
-0.6338    0.8967    0.4743
 0.7264    0.4353   -0.2558
 0.2660   -0.0802    0.8424
L = 3x3
 2.8437         0         0
  0   -4.6732         0
  0         0   -6.1705

```

```
eig(A)
```

```

ans = 3x1
 2.8437
-4.6732
-6.1705

```

b) Find the characteristic polynomial for this matrix.

```

syms x;
eq1=charpoly(A, x)

```

```
eq1 = x^3 + 8 x^2 - 2 x - 82
```

c) Replace x in the characteristic polynomial by the matrix A and evaluate the value. What is the value obtained? Try this with other square matrices.

```
Eq2=subs(eq1,A)
```

```
Eq2 =
(-31  -10  -31)
(-54  -82   11)
(-73  -46   3 )
```

3) Generate an integer matrix A of order 5x3. Find T_A and T_A^T . Evaluate the eigenvalues of T_A and T_A^T . Verify the result that “The non-zero eigenvalues of T_A and T_A^T are same”.

```
clc
```

```
A=randi([0,9],5,3);
rank(A);
A1=A*A'
```

```
A1 = 5×5
    133    110    125    120    39
    110     93     97     92    34
    125     97    138    138    31
    120     92    138    149    28
     39     34     31     28    13
```

```
A2=A'*A
```

```
A2 = 3×3
    267    151    180
    151     94     92
    180     92    165
```

```
% Compute Eigenvalues for both the matrices
E_A1=eig(A1)
```

```
E_A1 = 5×1
   -0.0000
    0.0000
    3.9405
   39.2287
  482.8308
```

```
E_A2=eig(A2)
```

```
E_A2 = 3×1
    3.9405
   39.2287
  482.8308
```

```
% Remove near-zero eigenvalues for comparison
E_A1_non_zero = E_A1(E_A1 > 1e-10)
```

```
E_A1_non_zero = 3×1
    3.9405
   39.2287
  482.8308
```

```
E_A2_non_zero = E_A2(E_A2 > 1e-10)
```

```
E_A2_non_zero = 3×1
    3.9405
   39.2287
  482.8308
```

```
% Compare non-zero eigenvalues
if all(abs(E_A1_non_zero - E_A2_non_zero) < 1e-10)
    disp('The non-zero eigenvalues of A*A' and A'*A are the same.');
```

```
else
    disp('The non-zero eigenvalues of A*A' and A'*A are NOT the same.');
```

```
end
```

The non-zero eigenvalues of A*A' and A'*A are the same.

4) Generate a matrix with eigenvalues as 3,5,7,11 and eigenvectors as [1,1,2,2], [1,0,1,0], [2,1,2,0] and [0,1,1,0].

```
clc

% Enter the given eigenvalues
Ld1 = 3;
Ld2 = 5;
Ld3 = 7;
Ld4= 11;

% Enter the given eigenvectors
Ev1 = [1;1;2;2];
Ev2 = [1;0;1;0];
Ev3 = [2;1;2;0];
Ev4=[0;1;1;0];

V = [Ev1,Ev2,Ev3,Ev4];
D = diag([Ld1,Ld2,Ld3,Ld4]);
A = V*D*inv(V)
```

```
A = 4x4
     9     4    -4    -1
    -4     7     4    -4
    -2     4     7    -5
     0     0     0     3
```

5) Using MATLAB generate a 5×5 matrix of rank 2.

a) Obtain a symmetric matrix $B = A + A'$ and find rank of B .

```
A = randi([0 9], 5, 2)*randi([0 9], 2,5);
B=A+A'
```

```
B = 5x5
    18    26    49    42    22
    26    20    63    93    72
    49    63   146   169   119
    42    93   169   120    65
    22    72   119    65    28
```

```
rankB=rank(B)
```

```
rankB =
     4
```

b) Find the eigenvalues and eigenvectors of B using MATLAB.

```
[eigenvectors, eigenvalues] = eig(B)
```

```
eigenvectors = 5x5
   -0.0586   -0.2646    0.3404    0.8819    0.1813
    0.4964    0.5742    0.5634   -0.0774    0.3170
    0.4879   -0.5038   -0.2928   -0.1364    0.6355
```

```

-0.4352    0.5098   -0.4357    0.1740    0.5749
-0.5681   -0.2942    0.5396   -0.4091    0.3637
eigenvalues = 5x5
-85.0661     0         0         0         0
     0   -1.5812     0         0         0
     0         0   -0.0000     0         0
     0         0     0    6.2221     0
     0         0     0     0  412.4252

```

c) Are their complex eigenvalues? (Check the same for more symmetric matrices). Explain why or why not?

```

% Check if there are complex eigenvalues
if isreal(eigenvalues)
    disp('All eigenvalues are real.');
```

```

else
    disp('There are complex eigenvalues.');
```

```

end

```

All eigenvalues are real.

d) How many eigenvalues of B are zero? Is there a relation between the number of zero eigenvalues of B and the rank of B?

```

% To count how many eigenvalues are zero
zeroEigenvalues = sum(abs(diag(eigenvalues)) < 1e-6)

```

```

zeroEigenvalues =
1

```

```

expected_Zero_Eigenvalues = size(B,1) - rankB

```

```

expected_Zero_Eigenvalues =
1

```

```

% To check whether there is a relation between the number of zero
eigenvalues of B.
if zeroEigenvalues == expected_Zero_Eigenvalues
    disp('The number of zero eigenvalues matches the expected relation with
the rank of B.');
```

```

else
    disp('The number of zero eigenvalues does not match the expected
relation with the rank of B.');
```

```

end

```

The number of zero eigenvalues matches the expected relation with the rank of B.

e) Verify if the eigenvectors of B are orthogonal.

```

% Check if eigenvectors are orthogonal.
orthogonalCheck = eigenvectors' * eigenvectors;

```

```

% Check if eigenvectors' * eigenvectors is close to identity matrix.

```



```

if norm(orthogonalCheck - eye(size(B,1))) < 1e-10
    disp('Eigenvectors are orthogonal. ');
else
    disp('Eigenvectors are not orthogonal. ');
end

```

Eigenvectors are orthogonal.

6) Using MATLAB generate a 10×5 matrix of rank 3.

a) Obtain a symmetric matrix $S_1 = A^T A$ and $S_2 = A A^T$

```

clc
A = randi([0 9], 10, 3)*randi([0 9], 3,5)

```

```

A = 10x5
    42    43    76    36    86
    51    73   111    37    95
    45    60   113    37    99
    48    58   106    40   102
    45    60    87    33    81
    69    79    97    51   113
    45    57   117    39   105
    63    68    79    47   101
    51    69    99    37    91
    72    87   120    54   126

```

```

S1 = A' * A

```

```

S1 = 5x5
    29259    35796    53565    22479    54051
    35796    44186    66494    27468    66370
    53565    66494   103211    41501   101481
    22479    27468    41501    17339    41803
    54051    66370   101481    41803   101339

```

```

S2 = A * A'

```

```

S2 = 10x10
    18081    23219    22904    22778    19236    25221 ...
    23219    30645    29992    29618    25248    32675
    22904    29992    29564    29196    24696    31880
    22778    29618    29196    28908    24444    31742
    19236    25248    24696    24444    20844    27120
    25221    32675    31880    31742    27120    35781
    23667    30861    30504    30138    25416    32811
    21952    28280    27580    27524    23520    31192
    21791    28641    28000    27706    23640    30743
    28665    37311    36492    36234    30888    40473

```

b) Find the rank of S_1 and S_2 .

```

rank(S1)

```

```

ans =
     3

```

```

rank(S2)

```

```
ans =  
3
```

c) Find the eigenvalues of S1 and S2. What do you notice about them? Explain.

```
EvalS1 = eig(S1)
```

```
EvalS1 = 5×1  
105 ×  
-0.0000  
0.0000  
0.0048  
0.0180  
2.9305
```

```
EvalS2 = eig(S2)
```

```
EvalS2 = 10×1  
105 ×  
-0.0000  
-0.0000  
-0.0000  
-0.0000  
0.0000  
0.0000  
0.0000  
0.0048  
0.0180  
2.9305
```

OBSERVATION: The symmetric matrices have same nonzero eigenvalues.

7) Using MATLAB generate a 7× 7 matrix A of rank 2.

a) Obtain a skew-symmetric matrix $B = A - A'$ and find rank of B.

```
clc
```

```
A = randi([0 9],7,2)*randi([0 9],2,7);  
B = A - A'
```

```
B = 7×7  
0    -31   -40   -90   -27   -38   -66  
31     0   -40   -51   -9   -23   -55  
40    40    0    -3    15    23    2  
90    51    3     0    30    10   -3  
27     9   -15   -30    0    -6   -24  
38    23   -23   -10    6     0   -25  
66    55   -2     3    24    25    0
```

```
rank(B)
```

```
ans =  
4
```

b) Find the eigenvalues and eigenvectors of B using MATLAB.

```
[eigenvectors, eigenvalues] = eig(B)
```

```
eigenvectors = 7×7 complex
-0.5571 + 0.0043i -0.5571 - 0.0043i 0.3129 - 0.0664i 0.3129 + 0.0664i ...
-0.3588 + 0.1707i -0.3588 - 0.1707i -0.3151 + 0.0698i -0.3151 - 0.0698i
0.0822 + 0.2473i 0.0822 - 0.2473i 0.3262 + 0.3415i 0.3262 - 0.3415i
0.0924 + 0.4465i 0.0924 - 0.4465i -0.0026 - 0.5168i -0.0026 + 0.5168i
-0.1553 + 0.1481i -0.1553 - 0.1481i 0.1044 - 0.0114i 0.1044 + 0.0114i
-0.0878 + 0.2176i -0.0878 - 0.2176i -0.4120 + 0.2358i -0.4120 - 0.2358i
0.1176 + 0.3756i 0.1176 - 0.3756i 0.1255 + 0.2264i 0.1255 - 0.2264i
eigenvalues = 7×7 complex
102 ×
0.0000 + 1.6591i 0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i ...
0.0000 + 0.0000i 0.0000 - 1.6591i 0.0000 + 0.0000i 0.0000 + 0.0000i
0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.2266i 0.0000 + 0.0000i
0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 - 0.2266i
0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i
0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i
0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i 0.0000 + 0.0000i
```

c) Are their real eigenvalues? (Check the same for more skew-symmetric matrices). Explain why or why not?

```
% Check if eigenvalues are real (they should be imaginary for skew-symmetric matrices).
```

```
if all(isreal(diag(eigenvalues)))
    disp('All eigenvalues are real.')
else
    disp('Not all eigenvalues are real. There are imaginary eigenvalues.')
end
```

Not all eigenvalues are real. There are imaginary eigenvalues.

d) Do you have one eigenvalue as zero? Can you explain why?

```
% Check if there s a one or zero eigenvalue.
has_zero_eigenvalue = any(abs(diag(eigenvalues)) < 1e-10)
```

```
has_zero_eigenvalue = logical
1
```

e) In a similar manner generate a skew-symmetric matrix from an 8× 8 matrix. Does C have a zero eigenvalue? Why or why not?

```
% Create an 8*8 matrix and generate a skew symmetric matrix C.
C = rand(8, 8) - rand(8, 8)'
```

```
C = 8×8
-0.0379 -0.0702 0.3955 -0.0678 0.4979 0.0236 0.1347 0.5349
0.8030 0.3393 0.7990 0.1776 -0.1368 -0.1909 -0.0899 -0.4045
0.8765 -0.8370 0.7782 0.3024 0.5770 0.2769 0.6892 -0.7163
-0.2779 -0.0291 -0.0559 0.0268 0.0755 0.6306 0.4374 -0.3569
0.2340 -0.0183 -0.4708 -0.3149 0.3459 -0.5785 0.1937 0.3936
-0.3884 -0.2197 -0.1934 -0.3858 0.0241 -0.0997 0.0255 -0.3515
0.2762 0.2477 0.0310 0.0211 0.0439 -0.0575 -0.1909 0.4623
-0.5027 -0.0091 0.2032 0.6056 0.4218 0.5154 0.3031 -0.3513
```

```
% Find the eigenvalues of C and check if it has a zero eigenvalue.
```

```
eigenvalues_C = eig(C)
```

```
eigenvalues_C = 8×1 complex  
    0.6645 + 0.8721i  
    0.6645 - 0.8721i  
    0.7945 + 0.0000i  
   -0.3090 + 0.8918i  
   -0.3090 - 0.8918i  
   -0.2334 + 0.3960i  
   -0.2334 - 0.3960i  
   -0.2282 + 0.0000i
```

```
has_zero_eigenvalue_C = any(abs(eigenvalues_C) < 1e-10)
```

```
has_zero_eigenvalue_C = logical  
    0
```

```
if has_zero_eigenvalue_C  
    disp('C has no zero eigenvalue.')  
else  
    disp('C has a zero eigenvalue.')  
end
```

```
C has a zero eigenvalue.
```

8) Generate a matrix that will transform the points on a circular membrane to an elliptic membrane with a stretch factor of 5 and 9 along the directions [1,1] and [1, -1] respectively.

```
clc  
% Eigenvector matrix for directions [1,1] and [1,-1]  
V = (1/sqrt(2)) * [1 1; 1 -1];  
  
% Diagonal scaling matrix with stretch factors.  
D = [5 0; 0 9];  
  
% Transformation matrix.  
T = V * D * inv(V)
```

```
T = 2×2  
    7    -2  
   -2     7
```